



Color Image Steganography Method Based on RGB Model and Edge Detection

Hsing-Han Liu¹ · Yu-Ting Su¹

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Abstract

Due to advancements in camera manufacturing technology, photography has evolved from the black-and-white photos of the past to today's high-definition color images. This shift has also led to the gradual transition of information hiding techniques from grayscale images to color images. In recent years, scholars have continuously proposed information hiding techniques for color images. Based on the RGB model and edge detection, this study combines pixel value differencing and edge detection for color image steganography. This study improves on previous color image pixel value differencing—which initially splits the image into red, green, and blue layers—and combines with edge detection to embed more confidential information while maintaining a good camouflage quality due to the complex and less detectable textures on the edges. First, after dividing the carrier image into blocks of different sizes, the Laplacian of the Gaussian is used to determine whether it is an edge block; if it is an edge block, more confidential information is embedded. The procedure first uses the red and green layers as one group for steganography before using the green layer (with confidential information) and the blue layer as one group for steganography. The two sets of green layers are averaged and the red and blue layers with confidential information are adjusted synchronously. Finally, the adjusted red, green, and blue layers are superimposed to complete the steganographic procedure of this study. Compared with previously proposed methods, using the Lean image as an example, concealing capacity can be improved by approximately 23%. Extensive testing with 450 images of the color image database revealed an increase in concealing capacity of 15%, with no reduction in image quality of PSNR value is 33.51 dB. Furthermore, this study uses RS detection and pixel difference histogram for security analysis to verify the security of the proposed method. The experimental results demonstrated that the method proposed resists the detection of RS and pixel difference histograms. This study presents an irreversible information hiding method, where the experimental process involves simultaneous computation of all three RGB layers. Consequently, it is not possible to extract a single layer for information hiding. Therefore, this method can only be applied to information hiding in color images and cannot be used with grayscale images.

Keywords RGB model · Edge detection · Laplacian of the Gaussian · Pixel value differencing · Steganography

Abbreviations

RGB	Red, Green, and Blue
RS	Regular, Singular
LoG	Laplacian of Gaussian
PVD	Pixel-Value Differencing
HSV	Hue, Saturation, and Brightness
CYM	Cyan, Yellow, and Magenta
HVS	Human Visual System
LSB	Least-Significant Bit
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index Measure

1 Introduction

Recently, steganography has grown in popularity because of digital developments. However, steganography can be applied in confidential information transmission, image authentication, copyright protection, access control, data embedding, and annotations. In the military, steganography is often used for transmitting military intelligence, with confidential data concealed in a medium, such as videos or audio. However, owing to the popularity of 3C products, smart mobile devices have become easily accessible. Cameras have evolved from black-and-white to high-definition color, prompting steganography to switch from grayscale to color images. Recently, scholars have continuously proposed color image steganography methods. Xie et al. [1] proposed dividing the color images into three layers—red, green, and blue (RGB)—and concealing varying amounts of confidential information based on the least significant bit replacement method of the human visual system, which can effectively increase the amount of confidential information and maintain the original image quality. In 2017, Prasad and Pal [2] proposed a steganography technique applicable only to color images. This method used original pixel value differencing to divide the color image into three layers of RGB; the pixel value difference (PVD) of the relative positions of each layer was applied to conceal confidential information. However, the error of human vision to colors was not considered, resulting in slight chromatic aberrations and distortions of the image. Although the amount of confidential information hidden was greater than that obtained in previous studies, the image quality was reduced.

The embedding efficiency is a critical characteristic of steganography techniques, directly impacting their security [3]. It is defined by Fridrich and Soukal [4] as the number of message bits embedded per change resulting from embedding. In image-based steganography, high embedding efficiency means fewer changes to cover pixels are needed for a given embedding rate. The concept of embedding efficiency was initially introduced by Crandall [5] and was later adopted by Westfeld [6] for embedding in the DCT domain. It has since been recognized as a crucial attribute of steganography schemes [7]. Techniques with high embedding efficiency are less likely to disrupt the statistical properties of the cover image, thereby enhancing steganographic security. In essence, steganography methods that achieve high embedding efficiency produce stego-images with minimal distortion.

Steganography either focuses on increasing the amount of confidential information concealed or maintaining reliable image quality. This study uses color space, human visual system model, and Laplacian of the Gaussian edge detection technology [8], combined with the improved pixel value differencing, through the characteristic of different perception

of human eyes for colors and the principle of edge blocks not being easily detectable, to increase the amount of confidential information concealed and maintain the quality of the camouflaged images by hiding more confidential information in the easily overlooked colors of the images and edge blocks. This study uses Python to implement and test the proposed steganography method's correctness. Meanwhile, image quality analysis is also conducted on camouflaged images to maintain the balance between the amount of confidential information concealed and image quality. Finally, a security analysis is performed on the image with confidential information to ensure the image can effectively resist the analysis and detection mechanisms for confidential information.

This study employs spatial domain and irreversible information hiding techniques; therefore, frequency domain and reversible information hiding techniques are not discussed. The information hiding method proposed in this study involves simultaneous computation of all three RGB layers in the experimental process. Consequently, it is not possible to extract a single layer for information hiding. Hence, this method can only be applied to information hiding in color images and cannot be used with grayscale images. This study utilizes the Laplacian of the Gaussian edge detection technique and the properties of the RGB color space to enhance the capacity for information hiding while maintaining the image quality of the steganographic image. Finally, the security of the image is analyzed using pixel difference histogram analysis and RS detection techniques. The results show that this study's method can effectively resist these steganalysis detection techniques.

2 Literature Review

2.1 Edge Detection

The edge denotes a large contrast in brightness in the image and may or may not correspond to the boundary of the object in the image. The edge's satisfactory characteristics effectively reduce the amount of data to be processed while retaining the shape information of the object [9]. There are two approaches for detecting edges: "search-based" and "zero-crossing-based." The search-based method typically calculates the edge with a gradient vector and expresses the edge as a first-order derivative (Sobel and Canny methods) [10, 11]. The zero-crossing-based method uses Gaussian filtering to calculate the edge. The edge is located by the zero-crossing point of the second-order derivative in the image, for instance: Laplacian of the Gaussian (LoG) [8]. Table 1 shows the comparison of Sobel, Canny, and LoG edge detection techniques.

2.2 Steganographic Scheme based on PVD

The PVD methods introduced by Wu and Tsai [12] determine the capacity primarily based on the difference values between adjacent pixels. In areas of the image with smooth texture, the hiding capacity is limited, as these areas exhibit minimal changes in pixel values. Conversely, regions with complex textures can accommodate more hidden information due to their higher degrees of pixel value variation.

In the PVD hiding method, the number of digits allocated for data hiding is determined by comparing the value differences between pairs of pixels with a predefined set of quantization values, as illustrated in Tables 2 and 3. Further adjustments are made based on the specific difference value encountered.

Table 1 Comparison of edge detection technologies

Edge detection technology	Soble[10]	Canny[11]	LoG[8]
Advantages	1. Strongly robust to noise 2. Large amount of calculation 3. Fast operating speed	1. Low error rate 2. Precise positioning 3. High resolution	1. Good edge continuity 2. Precise positioning 3. Weaker edges can be extracted 4. Edges are sharpened and non-directional
Disadvantages	1. Low positioning accuracy 2. Low resolution	Design is complex and the operation is slow Fewer edges It is easy to mistake noise as an edge	If noise suppression is high, edge accuracy is low

Table 2 Difference scale range chart

R_k	Different difference intervals (k)					
	1	2	3	4	5	6
l_k	0	8	16	32	64	128
u_k	7	15	31	63	127	255
n	3	3	4	5	6	7

Table 3 Definition table of variables in relevant formulas used in this study

No	Symbol	Description
1	P_0	The first pixel in the block, also called the benchmark pixel
2	P_R, P_G, P_B	The original pixel value of the R, G, B layer
3	d_1	The difference between R and G layers
4	d_2	The difference between G and B layers
5	n	Number of steganogram bits extracted as confidential information
6	b	Extracted decimal bits from the confidential information
7	d'_1	Difference value of the adjusted P_R and P_G
8	d'_2	Difference value of the adjusted P_{G1} and P_B
9	l_i, u_i	The lowest and highest value in the range of the interval
10	$P_{R1}, P_{G1}, P_{G2}, P_{B1}$	The pixel value of the first calculation of the R_1, G_1, G_2 and B_1 layer
11	m	The difference value between d and d'
12	P'_R, P'_G, P'_B	The stego-pixel value of the R, G, B layer

Step 1: Given a grayscale image with a height of M pixels and width of N pixels, the total number of pixels in the image is $M \times N$. Each pixel can be represented using an 8-digit binary system based on its light and dark characteristics. To divide the entire cover image into sub-image blocks of size 1×2 pixels, adjacent groups of

pixels are considered. This process results in a set of defined locations (i, j) and $(i, j+1)$, along with their corresponding pixel values $f(i, j)$ and $f(i, j+1)$. To ensure non-overlapping blocks, the cover image is scanned in a Z-type pattern.

Step 2: The difference value between two pixels, $f(i, j)$ and $f(i, j+1)$, is denoted as d . The range of d lies between 0 and 255, with larger values indicating blocks located in areas of the cover image with complex textures. Conversely, smaller values indicate blocks situated in regions with smoother textures.

Step 3: The range of difference values, d , is divided into six sub-ranges: 0 to 7, 8 to 15, 16 to 31, 32 to 63, 64 to 127, and 128 to 255, as depicted in Table 2. Each block has a specific number of suppressible data bits: 3, 3, 4, 5, 6, and 7, respectively. Let n represent the total number of digits that can be hidden. By extracting n digits from the ciphertext and converting them from binary to decimal, the resulting value, denoted as w , is added to the minimum value l_k . This calculation yields the new difference value as follows:

$$d' = \begin{cases} l_k + w & \text{if } d \geq 0 \\ -(l_k + w) & \text{if } d < 0 \end{cases} \quad (1)$$

Step 4: The adjustment of the pixel values $f(i, j)$ and $f(i, j+1)$ involves utilizing the new difference values, d' and d , to conceal the information. The new values of $f'(i, j)$ and $f'(i, j+1)$ are calculated based on the relationship between the pixel values in the following manner:

$$f'(i, j), f'(i, j+1) = \begin{cases} f(i, j) - \lceil \frac{d'-d}{2} \rceil, f(i, j+1) + \lceil \frac{d'-d}{2} \rceil, & \text{if } d \text{ is odd} \\ f(i, j) - \lceil \frac{d'-d}{2} \rceil, f(i, j+1) + \lceil \frac{d'-d}{2} \rceil, & \text{if } d \text{ is even} \end{cases} \quad (2)$$

Step 5: Repeat steps 1 to 4 for the next 1×2 block.

In this approach, the retrieval of confidential data is accomplished through the following steps.

Step 1: Obtain d' by using the known $f(i, j)$ and $f'(i, j+1)$ according to Eq. (3).

$$d' = |f'(i, j+1) - f(i, j)| \quad (3)$$

Step 2: Determine the minimum value l_k for the specific area and define the range of difference values for d' based on Table 2.

Step 3: Utilize Eq. (4) along with the known d' and l_k to calculate the decimal value of w .

$$w = \begin{cases} d' - l_k, & \text{if } d' \geq 0 \\ d' + l_k, & \text{if } d' < 0 \end{cases} \quad (4)$$

Step 4: Convert the obtained w into binary form to extract the hidden ciphertext message from the 1×2 image block.

2.3 Introduction to Color Images

As early as the 1840s, many scientists attempted to create color images. In 1855, Scottish physicists Maxwell, James Clerk first proposed the three-color method using three-color

light imaging based on the RGB color principle. This is the origin of Color Imaging Technology [13]. However, the first color photograph was actually taken in 1861 by the English photographer Thomas Sutton.

Color images are mainly composed of primary R (red), G (green), B (blue), and secondary C (cyan), Y (yellow), and M (magenta) colors. The color space or hue, saturation, value (HSV) refers to the way in which color is organized, based on the premise that the human eye perceives color mainly through three elements: Hue (hue), Saturation (saturation), and Value (brightness). Hue represents the color, saturation determines the shade of the color, whereas brightness affects the intensity of the color. Color images are actually the visual effects of light perceived by human eyes and brains in daily life. The interactions among colors can mainly be divided into two categories: “additive color” and “subtractive color”. Additive color is RGB, which when mixed in equal measures becomes white; subtractive color refers to CYM, which when mixed in equal measures becomes black.

As human vision has different degrees of perception to the three primary RGB colors, image processing can define another encoding mode which is similar to RGB, that is, YUV, where Y represents brightness, and U and V represent chroma and saturation, respectively. The color scale of RGB is 0–255, whereas the color scale is 16–235 for YUV. Their relations are shown in the formula.

$$\begin{aligned} Y &= 0.299R + 0.587G + 0.114B \\ U &= -0.169R - 0.331G + 0.500B \\ V &= 0.500R - 0.419G - 0.081B \end{aligned} \quad (5)$$

2.4 Human visual system

Gonzalez and Woods [14] propose that human vision is a sensory organ, depending on the human visual system (HVS) that produces different structures according to the needs of the individual. Therefore, it has different sensitivities to different wavelengths. The light passing through the pupil enters the inside of the eyeball, where the amount of light that enters is controlled by the cornea. Eventually the amount of pigment received by the iris generates the various colors seen by human eyes. The perception of the human eye mainly relies on the cone cells and rod cells in the retina to distinguish color and brightness. There are approximately 8 million cone cells, which can accept light ranging in wavelength from 400 to 700 nm, and are mainly responsible for color vision. According to the different light pigments, there are three types of receptors, namely red, green, and blue, with wavelengths of 564 nm, 534 nm, and 420 nm, respectively. In contrast, there are approximately 120 million rod cells which can accept light wavelengths from 400 to 600 nm, and are mainly responsible for perceiving light.

Therefore, the following points with HVS as the basis can be summarized:

1. The human eye is most sensitive to green light and least sensitive to blue light.
2. The HVS has different sensitivities to color at different levels of light.
3. The human eye is more sensitive to brightness than to color.
4. The human eye is less sensitive to noise on the edge of an image and more sensitive to noise in smooth blocks (within the image).

From the above points describing the characteristics of HVS, it is known that based on the degree of visual influence of human eyes on the color, brightness, and edge or smooth blocks of the image, ciphertext embedding using different amounts of concealment can be performed to improve the image quality of a stego-image (hidden image).

2.5 Steganography of color images

Owing to the development and evolution of information in recent years, color images have become routine to the general public. Therefore, academic research has gradually changed focus to the study of color images. The following sections will introduce scholars' studies on the steganography of color images in recent years.

2.5.1 Method proposed by Swain, G.

Two steganography methods were proposed by Swain, G. [15] in 2016, one with high density of information concealed, and the other a high security method. Both divide the color images into layers of the three primary RGB colors. For each layer, first create adjacent and non-repetitive 2×2 blocks and then create adjacent and overlapping 3×3 areas as blocks.

The steganography process for the 2×2 block method is as follows:

Step 1: Divide the original color image into three images: R, G, and B, and create adjacent and non-repetitive 2×2 blocks in each layer.

Step 2: Calculate the difference values: $d_1 = g_{x1} - g_r$, $d_2 = g_{x1} - g_b$, $d_3 = g_{x2} - g_r$, and $d_4 = g_{x2} - g_b$.

Step 3: Calculate the upper limit values l_1, l_2 and the lower limit values u_1, u_2 of g_{x1} and g_{x2} .

Step 4: Calculate the number of bits n_1, n_2 to be embedded into g_{x1} and g_{x2} .

$$n_1 = \text{floor}(\log_2 |u_1 - l_1 + 1|) \quad (6)$$

$$n_2 = \text{floor}(\log_2 |u_2 - l_2 + 1|) \quad (7)$$

Step 5: Convert the confidential information consisting of n_1, n_2 bits into decimal bits: b_1, b_2 , and calculate the stego-pixels: g_{x1}' , g_{x2}' .

$$g_{x1}' = \text{argmin}_e \{ |e - g_{x1}| | |e - g_r| \equiv b_1 \pmod{2^{n_1}}, e \in [l_1, u_1] \} \quad (8)$$

$$g_{x2}' = \text{argmin}_e \{ |e - g_{x2}| | |e - g_r| \equiv b_2 \pmod{2^{n_2}}, e \in [l_2, u_2] \} \quad (9)$$

The decryption process for the 2×2 block method is as follows:

Step 1: Divide the stego-color image into three images: R, G, and B, and create adjacent and non-repetitive 2×2 blocks in each layer.

Step 2: Calculate the differences: $d'_1 = g_{x1}' - g_r'$, $d'_2 = g_{x1}' - g_b'$, $d'_3 = g_{x2}' - g_r'$, and $d'_4 = g_{x2}' - g_b'$.

Step 3: From Step 3 in the steganography process, calculate the upper limit values l'_1, l'_2 and the lower limit values u'_1, u'_2 of g_{x1}' , g_{x2}' . Subsequently, use Step 4 in the steganography process to calculate n_1, n_2 .

Step 4: Convert n_1, n_2 into decimal bits b_1, b_2 .

$$b_1 \equiv |g_{x1}' - g_r'| \pmod{2^{n_1}} \quad (10)$$

$$b_2 \equiv |g_{x2}' - g_r'| \pmod{2^{n_2}} \quad (11)$$

The steganography process for the adjacent and overlapping 3×3 block method is as follows:

Step 1: Divide the original color image into three images, R, G, and B. Subsequently, in each layer, create 3×3 blocks which are adjacent and whose third row and the third column overlap with the first row and the first column of the next block. The central pixel is the standard pixel with value g_x , and the adjacent pixel values are g_u, g_l, g_r , and g_b .

Step 2: Calculate the difference values $d_1 = g_x - g_l$, $d_2 = g_r - g_x$, $d_3 = g_x - g_u$, and $d_4 = g_b - g_x$. If $|d_1| \leq T$, $|d_2| \leq T$, $|d_3| \leq T$, and $|d_4| \leq T$, the block will not be encrypted; skip directly to the next block. The ideal value of T is 10.

Step 3: If $|d_1| > T$ or $|d_2| > T$, calculate the upper limit l_h and lower limit u_h , otherwise, $l_h = 0$ and $u_h = 0$.

Step 4: If $|d_3| > T$ or $|d_4| > T$, calculate the upper limit l_v and lower limit u_v , otherwise, $l_v = 0$ and $u_v = 0$.

Step 5: Calculate the final upper limit l_f , lower limit u_f , the standard pixel value g_f , and standard value $flag$.

Step 6: Calculate the number of bits n to embed into g_x

$$n = \text{floor}(\log_2 |u_f - l_f + 1|) \quad (12)$$

Step 7: If $n > 0$, convert the confidential information into b decimal bits and embed it into the stego-pixel g'_x .

$$g'_x = \text{argmin}_e \{ |e - g_x| \mid |e - g_f| \equiv b \pmod{2^n} \} \quad (13)$$

Take g_{ul} as the indicator. If $flag = 0$, the least-significant bit (LSB) of g_{ul} is 0; otherwise, it is 1.

The decryption process is as follows:

Step 1: Divide the original color image into three images, R, G, and B. Subsequently, in each layer, create 3×3 blocks which are adjacent and whose third row and third column overlap with the first row and first column of the next block. The central pixel is the standard pixel with value g_x , and the values of adjacent pixels are g_u, g_l, g_r , and g_b .

Step 2: Calculate the difference values $d'_1 = g'_x - g'_l$, $d'_2 = g'_r - g'_x$, $d'_3 = g'_x - g'_u$, and $d'_4 = g'_b - g'_x$; if $d'_1 \leq T$, $d'_2 \leq T$, $d'_3 \leq T$, $d'_4 \leq T$, the block is not encrypted; execute the next block.

Step 3: Using Step 3 and Step 4 in the steganography process, determine l_h, u_h, l_v, u_v .

Step 4: If the LSB of g_{ul}' is 0, then $l_f = l_h$, $u_f = u_h$, and $g_f = g_l$; if the LSB of g_{ul}' is 1, then $l_f = l_v$, $u_f = u_v$, and $g_f = g_u$.

Step 5: Calculate the number of bits $n = \text{floor}(\log_2 |u_f - l_f + 1|)$ hidden in g'_x , and then convert it into decimal bits $b \equiv |g'_x - g_f| \pmod{2^n}$ to obtain the confidential information.

2.5.2 Method proposed by Prasad S. and Pal A. K.

In 2017, Prasad S. and Pal A.K [2], proposed an information hiding technique that can only be applied to color images. It used the original pixel value differencing (PVD) steganography method and improved it. The color images are divided into layers of the three primary RGB colors, and then pixels at the same position are used in each layer for the steganogram.

Step 1: Divide the original color image into three images: R, G, and B.

Step 2: Starting from the first pixel of each image, perform the operation in a zigzag manner: using (R, G) and (G, B) as a group to calculate the difference values between pairs: $d_1 = |R - G|$, $d_2 = |G - B|$. Referring to the interval scales (as shown in Table 2) for the original PVD provided in Sect. 2.2, determine the number of steganogram bits.

Step 3: Create the steganogram using the original PVD to obtain (R_1, G_1) , (G_2, B_1) .

Step 4: Calculate the pixels after creating the steganogram defined by $R' = R_1 - (G_1 - \frac{G_1 + G_2}{2})$, $G' = \frac{G_1 + G_2}{2}$, and $B' = B_1 - (G_2 - \frac{G_1 + G_2}{2})$.

The decryption process is as follows:

Step 1: Divide the stego-color image into three images: R, G, and B.

Step 2: Starting from the first pixel of each image, operate in a zigzag manner, using (R', G') and (G', B') as a group to calculate the difference values between each pair: $d'_1 = |R' - G'|$, $d'_2 = |G' - B'|$. Refer to the interval scale table of the original PVD to determine the number of steganogram bits.

Step 3: Decrypt the original PVD to obtain the confidential information.

2.5.3 Method proposed by Hassaballah, Hameed and Aly

Hassaballah, Hameed, and Aly [16] proposed a color image steganography technique based on Pixel-based Adaptive Directional (P-ADPVD) in 2020. Initially, it computes the gradient angle of a single layer and uses Histogram of Oriented Gradient (HOG) on 2×2 blocks in the angle-quantized image to identify edge pixels. Adaptive edge pixels (POI) are obtained by adjusting a threshold (T_h), while smooth region pixels in the image are termed PONI. Using POI and the original PVD method, three combinations of disguised pixels are calculated: (P_i, P_j) with (P_i, P_{j+1}) , (P_i, P_j) with (P_{i+1}, P_j) , and (P_i, P_j) with (P_{i+1}, P_{j+1}) . The pixel positions are illustrated in Fig. 1.

The embedding capacity (bits per pixel) of the three methods mentioned above does not exceed 3. Therefore, this study proposes an improved method.

In 2023, Abdulla [17] proposed performance evaluation criteria for image steganography. Performance metrics that define digital image steganography techniques include payload capacity, stego image quality, and security (detectability of the secret message). Increasing payload capacity tends to diminish both the quality of the stego image and its security. Conversely, achieving high stego image quality and security often requires compromising on payload capacity. Balancing these requirements has become essential yet increasingly challenging. Recent advancements in image steganography can be categorized into two approaches. The first approach focuses on embedding efficiency, where the secret message is hidden while minimizing distortion in the cover image. The second approach involves techniques based on distortion functions related to statistical detectability, where the secret message is concealed in specific parts of the cover image, such as textured or

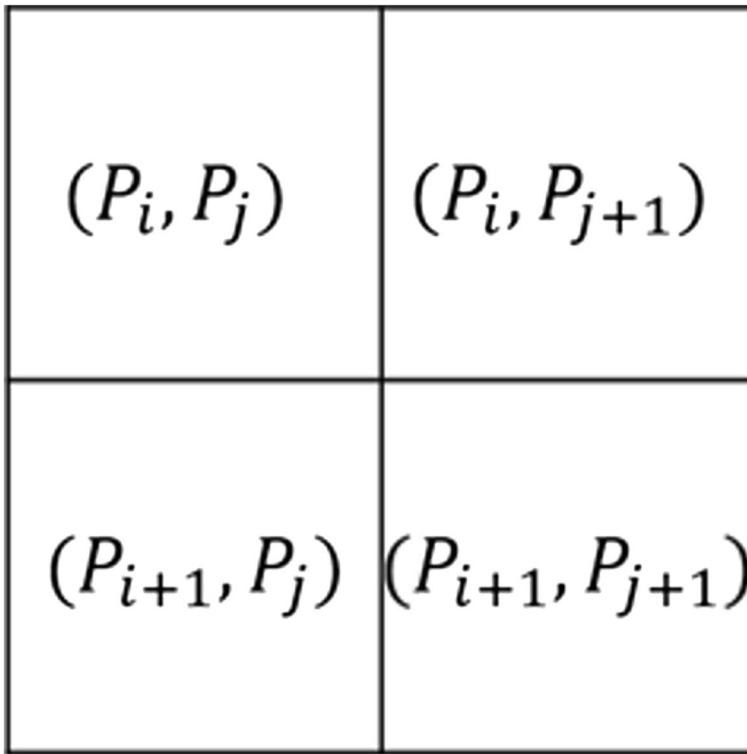


Fig. 1 Pixel positions

noisy regions. This study aims to provide researchers with insights for future work and guide them in designing efficient steganography techniques. It critically analyzes which of the two approaches can simultaneously meet all steganography requirements.

In 2021 and 2024, Khan et al. [18, 19] proposed techniques for image enhancement. Through image enhancement, the embedding efficiency of color image steganography techniques can be improved.

2.6 Evaluation Metrics for Information Hiding Techniques

The embedding capacity refers to the number of bits of confidential information embedded into the carrier image. The embedding capacity of the entire image is measured in bits, while the embedding capacity per pixel is measured in bpp (bits per pixel). The Peak Signal-to-Noise Ratio (PSNR) is used to measure image distortion, as shown in Eq. (14). The unit of PSNR is decibels (dB); the higher the value, the lower the distortion. A PSNR close to 50 dB indicates that the stego image has very little error. If the PSNR is greater than 30 dB, it is difficult for the human eye to detect the difference between the stego image and the original image. If the PSNR is between 20 and 30 dB, the human eye can easily perceive the difference between the images. If the PSNR is less than 20 dB, the original structure of the image can still be seen, but the image is noticeably distorted.

$$PSNR = 10 \times \log \frac{255^2}{MSE} \quad (14)$$

In Eq. (14), MSE stands for Mean-Square Error, which calculates the difference between the image before and after embedding, as shown in Eq. (15). A larger MSE indicates a greater error, while a smaller MSE indicates a smaller error. In these equations, W and H represent the width and height of the original image, x'_{ij} represents the stego pixel value, and x_{ij} represents the original pixel value.

$$MSE = \sum_{i=1}^{W \times H} \frac{(x'_{ij} - x_{ij})^2}{W \times H} \quad (15)$$

Wang et al. [20] proposed the Structural Similarity (SSIM) index in 2004, as shown in Eq. (16). SSIM is a measure of the similarity between two images, with values ranging from 0 to 1. A higher SSIM value indicates lower distortion and higher similarity between the two images. The computation is based on comparing factors such as luminance, contrast, and structure of the pixels in the images, as shown in Eq. (17)–(19). Therefore, SSIM provides a more accurate assessment of the similarity between images.

$$SSIM(x, y) = [l(x, y)]^\alpha [c(x, y)]^\beta [s(x, y)]^\gamma \quad (16)$$

$$l(x, y) = \frac{2u_x u_y + C_1}{u_x^2 + u_y^2 + C_1}, u_x = \frac{1}{N} \sum_{i=1}^N x_i \quad (17)$$

$$c(x, y) = \frac{2\sigma_x \sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}, \sigma_x = \left(\frac{1}{N-1} \sum_{i=1}^N (x_i - u_x)^2 \right)^{\frac{1}{2}} \quad (18)$$

$$s(x, y) = \frac{\sigma_{xy} + C_3}{\sigma_x \sigma_y + C_3}, \sigma_{xy} = \frac{1}{N-1} \sum_{i=1}^N ((x_i - u_x)(y_i - u_y)) \quad (19)$$

This study evaluates the effectiveness of the proposed method by comparing the PSNR and SSIM values between the stego image generated using the steganography technique and the cover image.

3 Method and Architecture

This study uses the RGB model and the technique of edge detection to increase the amount of concealed information and improve the quality of stego-images. The relevant variables are defined as shown in the table below. The procedures for embedding and extracting steganograms are introduced in sequence.

First, divide the colored carrier image into three layers: R, G, and B. Because the pixels in the smooth block of the G layer will have a large difference with those of the edge block, the G layer is used as the benchmark (as shown in Table 4), among which the coefficient of variation refers to the degree of dispersion. The larger the coefficient of variation is, the more

scattered the pixel values are and the greater the differences will be. In contrast, the smaller the coefficient of variation is, the more concentrated the pixel values will be near the average. The standard deviation refers to the degree of dispersion among individuals within a group. The larger the standard deviation is, the greater the difference between most of the pixel values and their average pixel values. In contrast, the smaller the standard deviation is, the closer the pixel values will be to the average. Subsequently, segment the carrier images into adjacent and non-repetitive 32×32 blocks. Use the LoG edge detection method to obtain the edge pixels so that whether a block is an edge block can be determined. Conceal the steganogram according to the different interval tables, and execute concealment of different components according to HVS so as to achieve the best concealment quality; if the carrier image cannot be divided into 32×32 blocks evenly, the remaining pixels will not carry steganographic information.

3.1 Embedding Procedure

The procedures for steganography concealment proposed in this study are as follows:

Step 1: Divide the colored carrier image into three layers: R, G, and B. Perform LoG edge detection on the G layer to generate an edge image. Segment both the G layer and the LoG edge image into adjacent and non-overlapping 32×32 blocks in a zigzag manner. If the carrier image cannot be divided into 32×32 blocks evenly, the remaining pixels will not be used, as shown in Figure 2.

Step 2: Set the first pixel in the block as the benchmark pixel P_0 , as shown in Figure 3.

Step 3: Use LoG edge detection to determine whether it is an edge block, where a pixel value greater than 64 is an edge pixel. If there are over 950 (inclusive) pixel values that are edge pixels in a block (excluding the benchmark pixel), the block is an edge block, as shown in Figure 4. Label the LSB in the benchmark pixel as “1.” If it is not an edge block, label the LSB in the benchmark pixel as “0.” However, neither the benchmark pixel nor pixels in its corresponding position will be used to embed confidential information.

Step 4: By denoting the upper limit of the interval as u_i and the lower limit as l_i , calculate the number of bits of confidential information that can be hidden as $n = \log_2(u_i - l_i + 1)$. According to the benchmark pixels in the block, whether it is an edge block can be known. If it is an edge block, refer to the edge interval scale table (as shown in Table 5) to hide the corresponding number of bits; if it is a non-edge block, refer to the modified interval scale (as shown in Table 6) to conceal the corresponding number of bits.

Step 5: Extract the R and G layers and verify whether the LSB of the benchmark pixel of the G layer is “1” or “0.” Calculate in a zigzag manner from the second pixel, starting with (P_R, P_G) as the first combination to calculate the difference $d_1 = |P_R - P_G|$. By

Table 4 Coefficient of variation and standard deviation for each layer

	Coefficient of variation	Standard deviation
R	1159.9422	34.0579
G	2796.0318	52.8775
B	2405.7822	49.0487

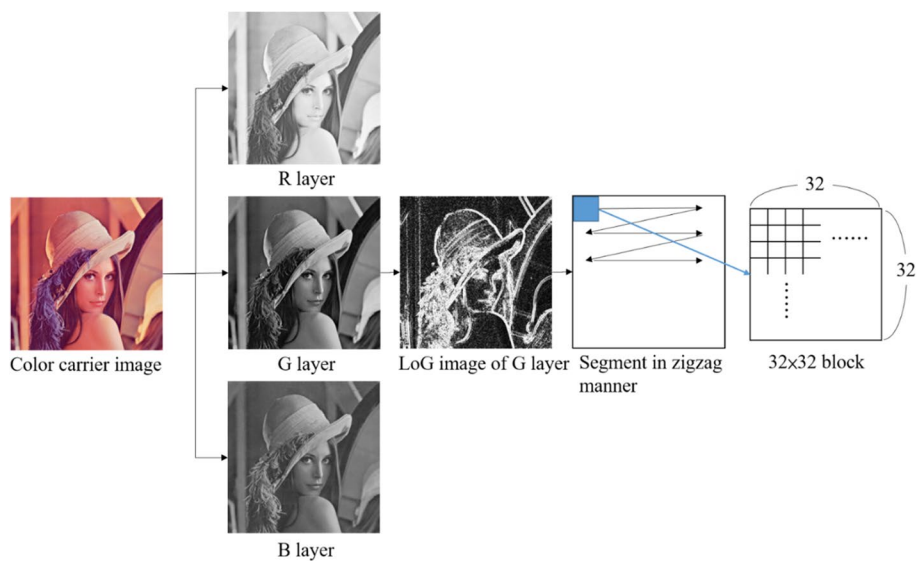


Fig. 2 Layer division and segmentation of the carrier image

P_0	P_1	P_2	...	P_{31}
P_{32}	P_{33}	P_{34}	...	P_{63}
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
P_{992}	...			P_{1023}

Fig. 3 Schematic diagram of benchmark pixels

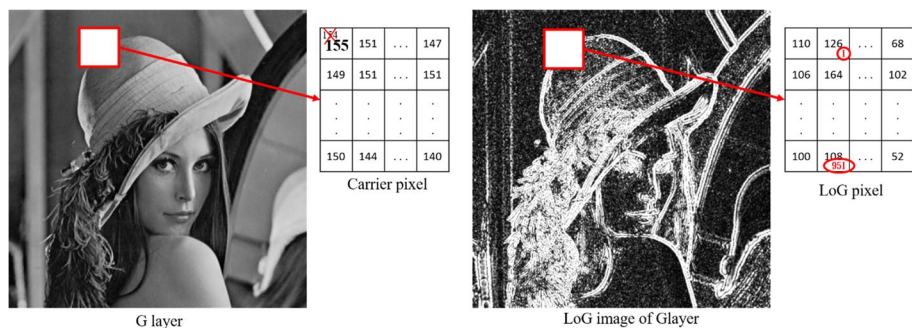


Fig. 4 Determining whether a block is an edge block in the image

Table 5 Edge interval scale table

Lowest value in the interval (l_i)	0	32	64	96	128
Highest value in the interval (u_i)	31	63	95	127	255
Maximum number of bits available for information concealment	5	5	5	5	7

Table 6 Modified interval scale table

Lowest value in the interval (l_i)	0	16	32	64	96	128	192
Lowest value in the interval (u_i)	15	31	63	95	127	191	255
Maximum number of bits available for information concealment	4	4	5	5	5	6	6

verifying the interval scale, the number of bits to conceal information is determined; subsequently, take n bits out from the confidential information and convert them to a decimal integer value b to calculate the new pixel difference d'_1 , as shown in Eq. (20).

$$d'_1 = \begin{cases} l_i + b, & \text{if } d \geq 0 \\ -(l_i + b), & \text{if } d < 0 \end{cases}, \text{ where } d = P_G - P_R \quad (20)$$

Step 6: The difference between the original pixel difference value d_1 and the new pixel difference value d'_1 is used to perform different proportions of concealing confidential information based on HVS. The ciphertext is distributed between two pixels proportionally to obtain the pixel (P_{R1}, P_{G1}) , as shown in Eq. (21).

$$(P_{R1}, P_{G1}) = \begin{cases} (P_R - \text{ceiling}_{m_R}, P_G + \text{floor}_{m_G}), & \text{if } d_1 \text{ is odd} \\ (P_R - \text{floor}_{m_R}, P_G + \text{ceiling}_{m_G}), & \text{if } d_1 \text{ is even} \end{cases} \quad (21)$$

where $m = d'_1 - d_1$, $\text{ceiling}_{m_R} = \lceil m \times 0.6 \rceil$, $\text{floor}_{m_R} = \lfloor m \times 0.6 \rfloor$, $\text{ceiling}_{m_G} = \lceil m \times 0.4 \rceil$, $\text{floor}_{m_G} = \lfloor m \times 0.4 \rfloor$

Step 7: Extract the layer of P_{G1} generated in Step 6 and the B layer of the original carrier and verify whether the LSB of the benchmark pixel of the G layer is “1” or “0.” Calculate in a zigzag manner from the second pixel, starting with (P_B, P_{G1}) as the first

group to calculate the difference $d_2 = |P_B - P_{G1}|$. Corresponding to the interval scale, the number of bits for concealing information is determined. Subsequently, take n bits out from the confidential information and convert them to a decimal integer value b to calculate the new pixel difference d'_2 , as shown in Eq. (22).

$$d'_2 = \begin{cases} l_i + b, & \text{if } d \geq 0 \\ -(l_i + b), & \text{if } d < 0 \end{cases}, \text{ where } d = P_{G1} - P_B \quad (22)$$

Step 8: The difference between the original pixel difference value d_2 and the new pixel difference value d'_2 is used to perform different proportions of concealing confidential information according to HVS. The ciphertext is distributed between two pixels proportionally to obtain the pixel (P_{B1}, P_{G2}) , as shown in Eq. (23).

$$(P_{B1}, P_{G2}) = \begin{cases} (P_B - \text{ceiling}_{m_B}, P_{G1} + \text{floor}_{m_{G1}}), & \text{if } d_2 \text{ is odd} \\ (P_B - \text{floor}_{m_B}, P_{G1} + \text{ceiling}_{m_{G1}}), & \text{if } d_2 \text{ is even} \end{cases} \quad (23)$$

where $m = d'_2 - d_2$, $\text{ceiling}_{m_B} = \lceil m \times 0.6 \rceil$, $\text{floor}_{m_B} = \lfloor m \times 0.6 \rfloor$, $\text{ceiling}_{m_{G1}} = \lceil m \times 0.4 \rceil$, $\text{floor}_{m_{G1}} = \lfloor m \times 0.4 \rfloor$,

Step 9: Calculate the camouflage pixel value in Eq. (24).

$$\begin{aligned} P'_G &= \text{round}\left(\frac{P_{G1} + P_{G2}}{2}\right) \\ P'_R &= P_{R1} - (P_{G1} - P'_G) \\ P'_B &= P_{B1} - (P_{G2} - P'_G) \end{aligned} \quad (24)$$

Step 10: Verify whether there is an overflow in the pixels. When the calculated pixel falls outside $[0-255]$, if P'_R exceeds 255, then $P'_R = 255$, P'_G, P'_B subtract the value of $(P'_R - 255)$, respectively. Same for the other two; if P'_R is below 0, then add the value of $(0 - P'_R)$ to $P'_R = 0$ and P'_G, P'_B , respectively. Same for the other two.

Step 11: Repeat Steps 4 to 10 to embed confidential text until the carrier images have completed the process.

3.2 Data Extraction Procedure

The data extraction procedure for the proposed method is described below.

Step 1: Divide the color carrier image into three layers, R, G, and B, and segment the G layer into adjacent and non-repetitive 32×32 blocks in a zigzag manner. Determine whether they are edge blocks according to the benchmark pixel.

Step 2: After determining whether it is an edge block, decide whether the edge interval scale table or the improved interval scale table used in this study should be used.

Step 3: The first pixel does not include confidential information, therefore, start from the second pixel. Taking (P'_R, P'_G) and (P'_G, P'_B) as a group, calculate the difference between pairs $d'_1 = |P'_R - P'_G|$, $d'_2 = |P'_G - P'_B|$ to correspond to the maximum number of bits available for concealment based on the interval scale table.

Step 4: Subtract the minimum value of the interval from d'_1, d'_2 to obtain confidential information.

3.3 Description of embedding and extraction example

The detailed explanation of the steganography process in this study is as follows.

Step 1: The color cover image is divided into R, G, and B layers, and the G layer is used to perform LoG edge detection to generate an edge image. Both the G layer and the LoG edge image are then divided into 32×32 non-overlapping blocks in a zigzag manner.

Step 2: Set the first pixel in the block as the benchmark pixel.

Step 3: Since the block contains 951 edge pixel values (excluding the reference pixel), the block is classified as an edge block. The LSB of the reference pixel is marked as "1". However, pixels in positions relative to the reference pixel do not embed any confidential information.

Step 4: As shown in Figure 5, assume that the second pixel of the R, G, and B layers are $P_R = 235$, $P_G = 154$, and $P_B = 104$, respectively, and the confidential message $s = 01101011010\dots$

1. First, calculate the difference between the R and G layers: $d_1 = |P_R - P_G| = |235 - 154| = 81$. According to Table 5, the number of embedded bits is 5 bits. Extract the corresponding 5 bits from the confidential message, which converts to the decimal value $b_1 = 13$. Calculate the new pixel difference $d'_1 = -(64 + 13) = -77$. Then, using the difference between the original difference d and the new difference d' , substitute into the formula to obtain the stego pixel values: $P_R = 235 - [4 \times 0.6] = 232$, $P_{G1} = 154 + [4 \times 0.4] = 155$.
2. Next, calculate the difference between the G1 and B layers: $d_2 = |P_{G1} - P_B| = |155 - 104| = 51$. According to Table 5, the number of embedded bits is 5 bits. Extract the corresponding 5 bits from the confidential message, which converts to the decimal value $b_2 = 13$. Calculate the new pixel difference $d'_2 = -(32 + 13) = -45$. Then, using the difference between the original difference d and the new difference d' , substitute into the formula to obtain the stego pixel values: $P_{G2} = 155 - [6 \times 0.4] = 152$, $P_B = 104 + [6 \times 0.6] = 107$.
3. Finally, calculate the stego pixels: $P'_G = \text{round}\left(\frac{155+152}{2}\right) = 154$, $P'_R = 232 - (155 - P'_G) = 231$, $P'_B = 107 - (152 - P'_G) = 109$.

The detailed explanation of the extraction process in this study is as follows.

Step 1: The color cover image is divided into R, G, and B layers. The G layer is then segmented into 32×32 non-overlapping blocks in a zigzag manner. Based on the reference pixel's LSB being 1, it is determined to be an edge block.

Step 2: The first pixel does not contain confidential information. Starting from the second pixel, calculate the differences in pairs: $d'_1 = |231 - 154| = 77$, $d'_2 = |154 - 109| = 45$. It is determined that the number of embedded bits is 5 bits for both. Subtracting the minimum value of the interval from d'_1 and d'_2 reveals the confidential information: $s_1 = 77 - 64 = 13 = (01101)_2$, $s_2 = 45 - 32 = 13 = (01101)_2$, as shown in Fig. 6.

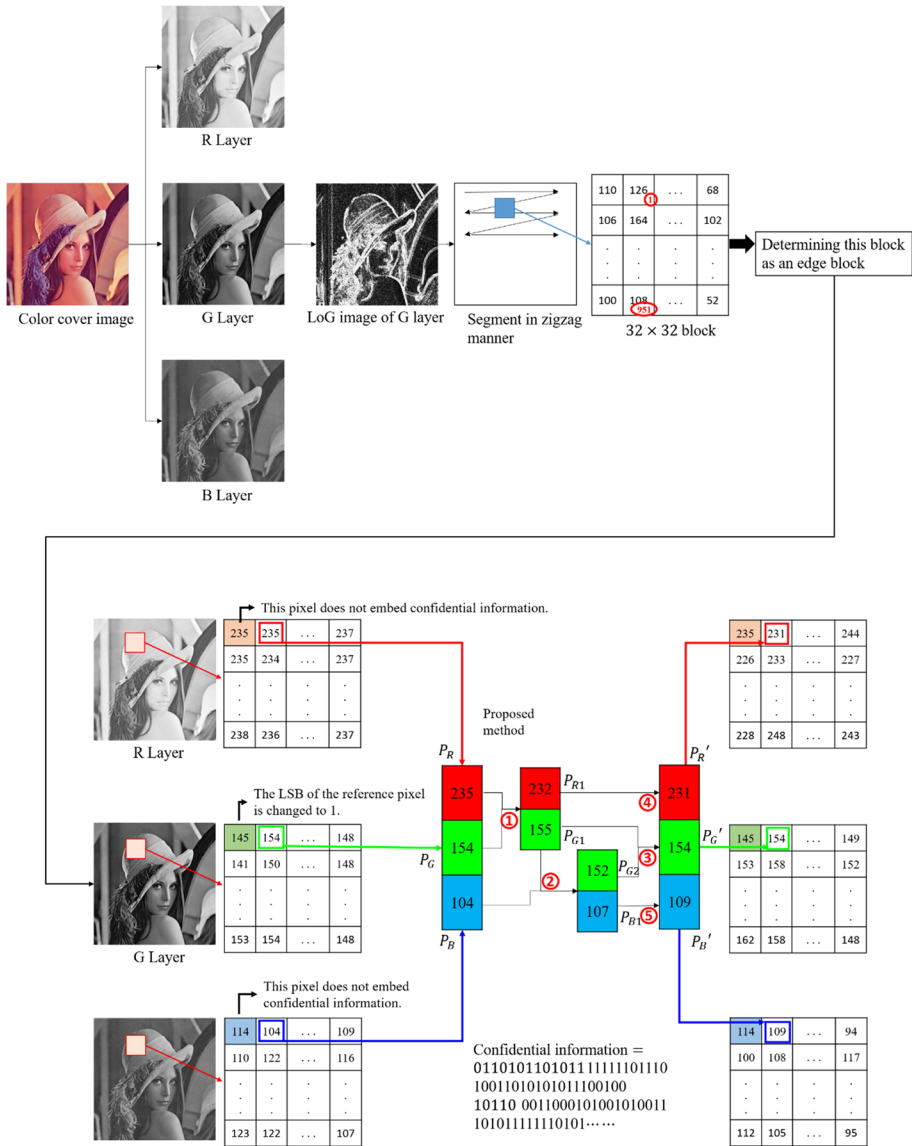


Fig. 5 Description of embedding example

4 Experimental Results

The LoG edge detection and PVD steganography were applied in this study to determine whether a block is an edge block using blocks of different sizes: 4×4 , 8×8 , 16×16 , and 32×32 . Furthermore, the color images of 512×512 pixels of Lena, Baboon, and Airplane, which are widely used in steganography experiments, were applied to compare

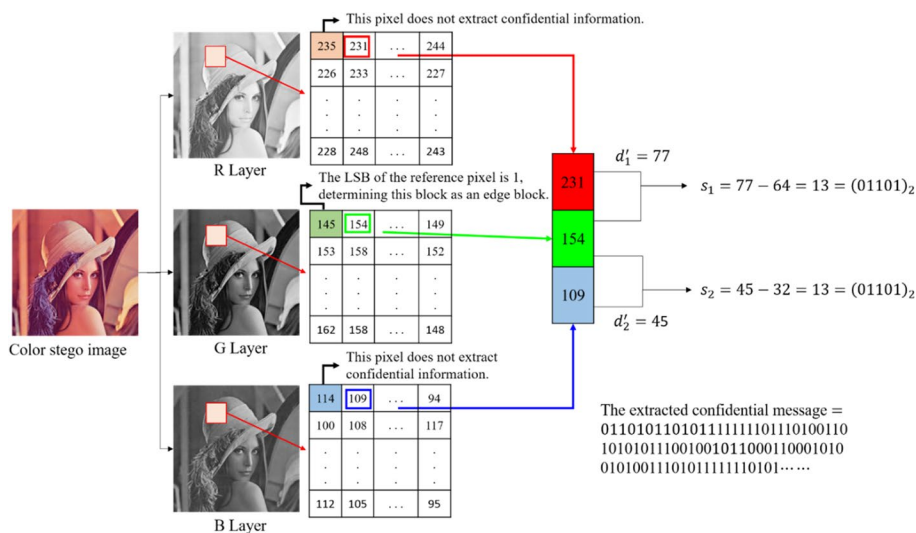


Fig. 6 Description of extraction example

the amount of confidential information and the quality of camouflaged images. After selecting the optimal scheme and comparing the amount of confidential information concealed and the quality of camouflaged images with the results from other scholars, the experiment finally verified that this method could effectively resist the detection of steganography analysis methods.

Given an image I with dimensions $M \times N$ (height M and width N), and we want to divide it into blocks of sizes $m \times n$, where m and n are the height and width of each block respectively. The process can be represented by the Eq. (25) and Table 7.

$$I = \{I_{m,n} \mid 1 \leq M \leq 255, 1 \leq N \leq 255\} \quad (25)$$

Since the cover image is a 512×512 two-dimensional array, the algorithm proposed in this study requires two nested for loops to control the row and column indices. Therefore, the time complexity of the steganography algorithm is $O(n^2)$, where n is the number of rows (or columns) in the image.

Table 7 Equation for dividing an image into blocks of different sizes

Blocks of different sizes	Block size	Divide the image into blocks
4×4	$B_{i,j} = \{B_{i,j}^{p,q} \mid 1 \leq p \leq 4, 1 \leq q \leq 4\}$	$I = \bigcup_{k=0}^{\lceil M/m \rceil - 1} \bigcup_{l=0}^{\lceil N/n \rceil - 1} B_{k,l}, m=4, n=4$
8×8	$B_{i,j} = \{B_{i,j}^{p,q} \mid 1 \leq p \leq 8, 1 \leq q \leq 8\}$	$I = \bigcup_{k=0}^{\lceil M/m \rceil - 1} \bigcup_{l=0}^{\lceil N/n \rceil - 1} B_{k,l}, m=8, n=8$
16×16	$B_{i,j} = \{B_{i,j}^{p,q} \mid 1 \leq p \leq 16, 1 \leq q \leq 16\}$	$I = \bigcup_{k=0}^{\lceil M/m \rceil - 1} \bigcup_{l=0}^{\lceil N/n \rceil - 1} B_{k,l}, m=16, n=16$
32×32	$B_{i,j} = \{B_{i,j}^{p,q} \mid 1 \leq p \leq 32, 1 \leq q \leq 32\}$	$I = \bigcup_{k=0}^{\lceil M/m \rceil - 1} \bigcup_{l=0}^{\lceil N/n \rceil - 1} B_{k,l}, m=32, n=32$

4.1 Experimental Environment

The hardware environment used in this experiment was Lenovo IdeaPad 330 laptop with CPU Intel Core i5-8250U 1.60 GHz, 12.0 GB RAM, and Windows 11 Home Edition operating system. The software environment used was the Spyder application program in Anaconda 3 which was written in Python. The above-mentioned environments were used to perform calculations of various steganography techniques and image qualities.

In addition to using the color images which have been widely used in steganography techniques for testing and data comparison, another 450 512×512 standard color images [21–23] from color image datasets were also used for testing. The embedded confidential information was generated with a sequence of 0 or 1 by a built-in random number generator in Python. The experiment yielded results in terms of hidden information, peak signal-to-noise ratio (PSNR), and Structural Similarity Index Measure (SSIM) of the images.

4.2 Experimental results of Lena, Baboon, and Airplane

For the three standard color images with 512×512 pixels of Lena, Baboon, and Airplane, the edge blocks were detected with different sizes of 4×4 , 8×8 , 16×16 , and 32×32 for the comparison of the amount of confidential information concealed and the quality of images. The experimental results are detailed below. Using Lena as an example, Table 8 shows that dividing into 32×32 blocks can conceal up to 2,427,406 bits of confidential information and achieve 31.46 dB PSNR and 0.80 SSIM.

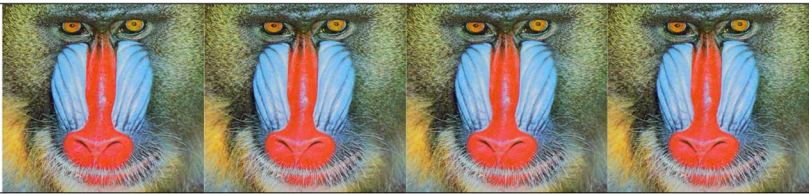
Using the Baboon color image as an example, experimental results indicate that after dividing the image into several 32×32 blocks, the amount of confidential information concealed can be as high as 2,326,637 bits, with PSNR reaching 31.26 dB, and SSIM 0.90, as shown in Table 9.

Using the Airplane color image as an example, the experimental results indicate that after dividing the image into several 32×32 blocks, the amount of confidential information concealed can be as high as 2,168,341 bits, and the best PSNR can reach 34.26 dB, with SSIM being 0.88, as shown in Table 10.

Table 8 Experimental results of Lena

Lena	4x4 blocks	8x8 blocks	16x16 blocks	32x32 blocks
Stego Image				
Experimental Results	Capacity : 2,306,763 bits (2.93 bpp) PSNR : 31.32 dB SSIM : 0.80	Capacity : 2,393,657 bits (3.04 bpp) PSNR : 31.45 dB SSIM : 0.80	Capacity : 2,418,952 bits (3.07 bpp) PSNR : 31.41 dB SSIM : 0.80	Capacity : 2,427,406 bits (3.09 bpp) PSNR : 31.46 dB SSIM : 0.80

Table 9 Experimental results of Baboon

Baboon	4×4 blocks	8×8 blocks	16×16 blocks	32×32 blocks
Stego Image				
Experimental Results	Capacity : 2,368,202 bits (3.01 bpp) PSNR : 29.26 dB SSIM : 0.89	Capacity : 2,320,952 bits (2.95 bpp) PSNR : 31.03 dB SSIM : 0.89	Capacity : 2,320,994 bits (2.95 bpp) PSNR : 31.29 dB SSIM : 0.90	Capacity : 2,326,637 bits (2.96 bpp) PSNR : 31.26 dB SSIM : 0.90

4.3 Comprehensive analysis of color image database

The color images were acquired from the three online databases, SIPI [21], NRCS [22], and TAMPERE17 [23]. Four hundred and fifty full-color lossless high-definition images with 512×512 pixels were used. Images with complex textures and smooth intervals were included. Images also include landscape, mountains, rivers, grasslands, forests, animals, cars, bicycles, utility poles, people, fruit, tables, chairs, books, bookshelves, sidewalks, aerial photographs, and buildings (as shown in Fig. 7). Thus, these images can be used to represent the content of most images.

The following experiments were conducted with 4×4 , 8×8 , 16×16 , and 32×32 blocks respectively. The results are shown in Table 11. The amounts of confidential information concealed were all in the range of 2,190,000 bits (2.80 bpp) and the PSNRs were all above 30 dB. Nonetheless, the 32×32 blocks yielded the best results. Therefore, this study used this method to conduct follow-up experiments.

4.4 Experimental Comparison with Other Methods

This section compares the method proposed in this study with the steganography techniques proposed by scholars such as Wu & Tsai [12], Swain [15], Prasad & Pal [2], and

Table 10 Experimental results of Airplane

Airplane	4×4 blocks	8×8 blocks	16×16 blocks	32×32 blocks
Stego Image				
Experimental Results	Capacity : 2,115,591 bits (2.69 bpp) PSNR : 33.12 dB SSIM : 0.87	Capacity : 2,142,488 bits (2.72 bpp) PSNR : 34.21 dB SSIM : 0.88	Capacity : 2,154,258 bits (2.74 bpp) PSNR : 34.26 dB SSIM : 0.88	Capacity : 2,168,341 bits (2.76 bpp) PSNR : 34.26 dB SSIM : 0.88

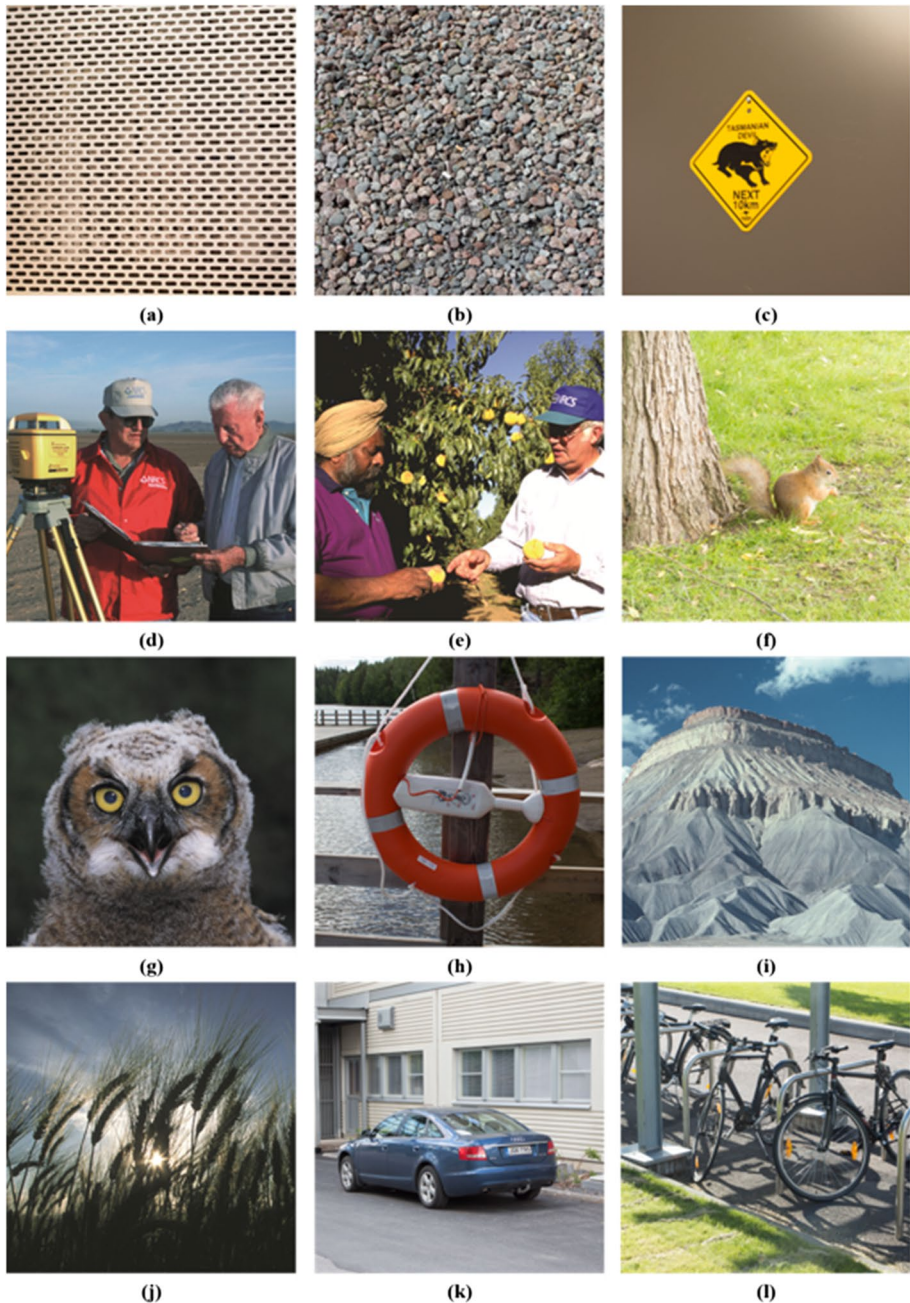


Fig. 7 Example images of the color image database

Hassaballah [16]. The carrier images used in the experiment were six 512×512 full-color lossless high-definition images of Lena, Peppers, Baboon, Airplane, Boat, and House (as shown in Fig. 8). The evaluation criteria included the amount of confidential information concealed (as shown in Table 12) and image quality (as shown in Table 13).

Table 11 Results of the comprehensive experiment of this study

blocks	4×4	8×8	16×16	32×32
capacity	2,196,048	2,185,270	2,195,307	2,198,830
bpp	2.79	2.78	2.79	2.80
PSNR	31.76	33.21	33.46	33.51

Table 12 shows the comparison of the amount of confidential information concealed (unit: bit) and bpp (the data in brackets of the table represent the amount of confidential information of each pixel) of each image by steganography techniques used by various scholars. The methods of steganography proposed by scholars such as Wu & Tsai, Swain, and Hassaballah all used each layer to hide confidential information, resulting in the amounts of confidential information concealed being relatively low, all below 2.00 bpp; in contrast, Prasad & Pal and this study both used the pixel of the corresponding positions of each layer to conceal the confidential information. As such, more confidential information can be concealed. In addition, this study also combined the characteristics of edge detection. Thus, the amount of confidential information concealed was even greater. The experimental results demonstrate that the minimum amount of concealed confidential information in this study is still 2.76 bpp, and the maximum is as high as 3.13 bpp, which is at least approximately 2% higher than that of Prasad & Pal. Furthermore, the average amount of hidden information has also increased by approximately 18%. These demonstrate that the proposed method in this study has superior experimental results among steganography techniques.

Table 13 shows the comparison of the PSNR (the value not in the brackets in the table is PSNR and its unit is dB) and SSIM (the value in brackets) of the images with concealed

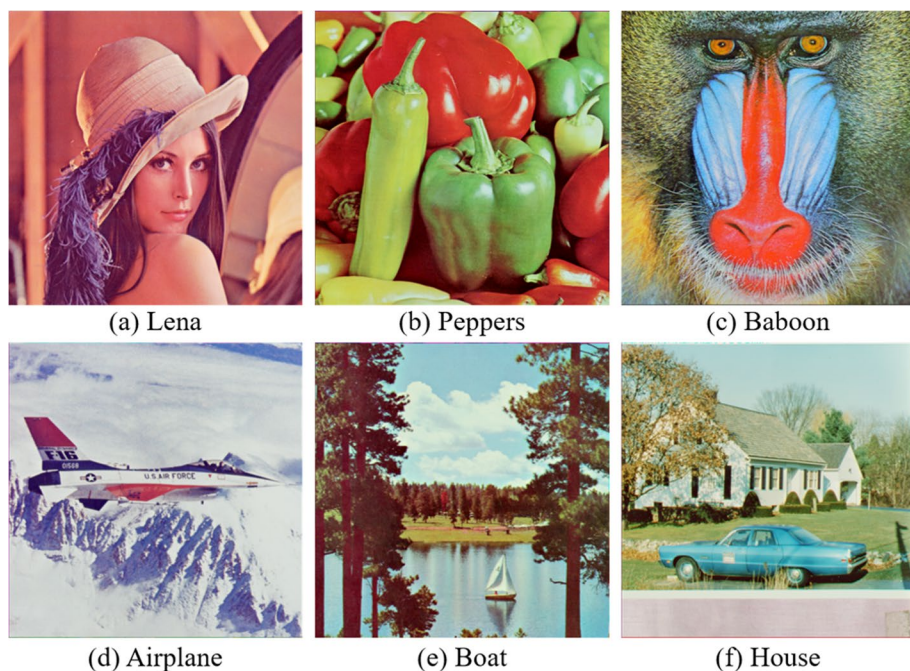
**Fig. 8** Experimental cover images

Table 12 Embedding capacities (in bits) and bpp of several steganographic methods

	Wu & Tsai [7]	Swain [10]	Prasad & Pal [2]	Hassaballah et al [14]	Proposed Method
Lena	1,234,394 (1.57)	1,341,192 (1.71)	1,976,671 (2.51)	1,548,888 (1.97)	2,427,406 (3.09)
Peppers	1,236,715 (1.57)	1,489,945 (1.89)	2,219,715 (2.82)	1,572,864 (2.00)	2,458,051 (3.13)
Baboon	1,405,795 (1.79)	1,267,690 (1.61)	1,753,707 (2.23)	1,571,784 (2.00)	2,326,637 (2.96)
Airplane	1,224,178 (1.56)	1,424,967 (1.81)	2,130,772 (2.71)	1,570,800 (2.00)	2,168,341 (2.76)
Boat	1,289,359 (1.64)	1,350,251 (1.72)	1,783,210 (2.27)	1,502,864 (1.91)	2,310,183 (2.94)
House	1,263,038 (1.61)	1,339,985 (1.70)	2,079,088 (2.64)	1,562,160 (1.97)	2,297,358 (2.92)
Average	1,275,766 (1.62)	1,369,005 (1.74)	1,990,527 (2.53)	1,554,893 (1.99)	2,353,300 (3.00)

Table 13 PSNR and SSIM comparison among several steganographic methods

	Wu & Tsai [7]	Swain [10]	Prasad & Pal [2]	Hassaballah et al [14]	Proposed Method
Lena	41.27 (0.97)	46.17 (0.99)	31.01 (0.74)	46.76 (0.99)	31.46 (0.80)
Peppers	40.83 (0.89)	48.49 (0.99)	32.29 (0.74)	43.75 (0.99)	29.73 (0.74)
Baboon	37.84 (0.98)	46.18 (0.99)	35.66 (0.88)	46.71 (0.99)	31.26 (0.91)
Airplane	40.64 (0.96)	47.29 (0.99)	33.11 (0.92)	45.54 (0.99)	34.26 (0.88)
Boat	39.74 (0.96)	47.06 (0.99)	30.10 (0.85)	45.58 (0.99)	32.54 (0.88)
House	39.94 (0.97)	44.73 (0.99)	34.59 (0.87)	44.74 (0.99)	32.20 (0.88)
Average	40.05 (0.96)	46.65 (0.99)	32.79 (0.83)	45.51 (0.99)	31.70 (0.86)

information after each scholar's steganography technique has been applied to them. Wu & Tsai, Swain, and Hassaballah's methods hid relatively low amounts of confidential information while maintaining high image quality and SSIMs. Prasad & Pal and the steganography method proposed in this study had relatively inferior image quality since our focus was on concealing a large amount of confidential information. Nonetheless, the PSNRs were still within an acceptable range, with an average of 31.70 dB.

The steganography technique proposed in this study is compared with two watermarking techniques applied to color images [24, 25] (as shown in Table 14). However, due to the fundamental differences between steganography and watermarking techniques, only PSNR is compared. According to the comparison results in Table 14, the PSNR of the steganography technique proposed in this study is inferior to that of the two watermarking techniques applied to color images.

4.5 Security Analysis

A successful steganography technique should possess both security and undetectability. Traditionally, the human visual system has been relied upon to identify whether a carrier image contains concealed information. However, this approach is highly subjective and often ineffective in detection. Consequently, researchers have explored different security analysis techniques for various steganography methods, including RS detection [26] and pixel difference histogram analysis [27, 28]. In the following experiments, the camouflaged images generated by this study are subjected to RS detection and pixel difference histogram analysis to assess the effectiveness of our proposed method in resisting detection by these two techniques.

4.5.1 RS Detection

The RS detection method, proposed by Fridrich et al. [26], is effective not only for randomly arranged embedding positions but also for estimating the embedding capacity. In fact, in the LSB positions of the cover image, what appears to be noise in the LSBs actually has nonlinear correlations. Spatial domain LSB hiding techniques disrupt this correlation, making it possible to detect hidden messages within the stego image using this correlation. Figure 9 illustrates the security of the proposed method against statistical RS detection technology [26]. The results for the analysis of the Lena cover image using RS detection technology are depicted in Fig. 9(a). For the analyses using 1-bit LSB and 3-bit LSB stego-images, Figs. 9(b) and (c) respectively display the results. In Fig. 9(d)–(f), the analysis results are shown for the R, G, and B layers of the stego-image generated using the proposed method.

The RS detection results, as shown in Figs. 9(a)–(f), indicate the evaluated embedding rates as follows: -0.01, 0.94, 0.89, 0, -0.07, and 0.06, respectively. These results demonstrate that RS detection technology is effective in detecting LSB steganography but lacks effectiveness in detecting the stego-images generated using the proposed method. Thus, the combination of the RGB model and edge detection in the steganography method proposed in this study proves to be both effective and robust against RS detection technology.

4.5.2 Pixel Difference Histogram Analysis

Zhang et al. [27] mentioned the Pixel Difference Histogram (PDH) analysis in 2004. This analysis is based on the principle that the histogram of pixel differences will exhibit an unnatural staircase-like distribution after secret messages are embedded using the PVD steganography method. This detection technique was developed to identify stego images created using the PVD steganography technique. During a PDH analysis, the frequency of

Table 14 PSNR comparison with two watermarking techniques

	Duan et al. [21] (2020)	Duan et al. [22] (2023)	Proposed Method
Lena	45.46	43.75	31.46
Peppers	45.73	43.30	29.73
Baboon	44.44	40.51	31.26
Airplane	45.29	45.69	34.26
Average	45.23	43.31	31.70

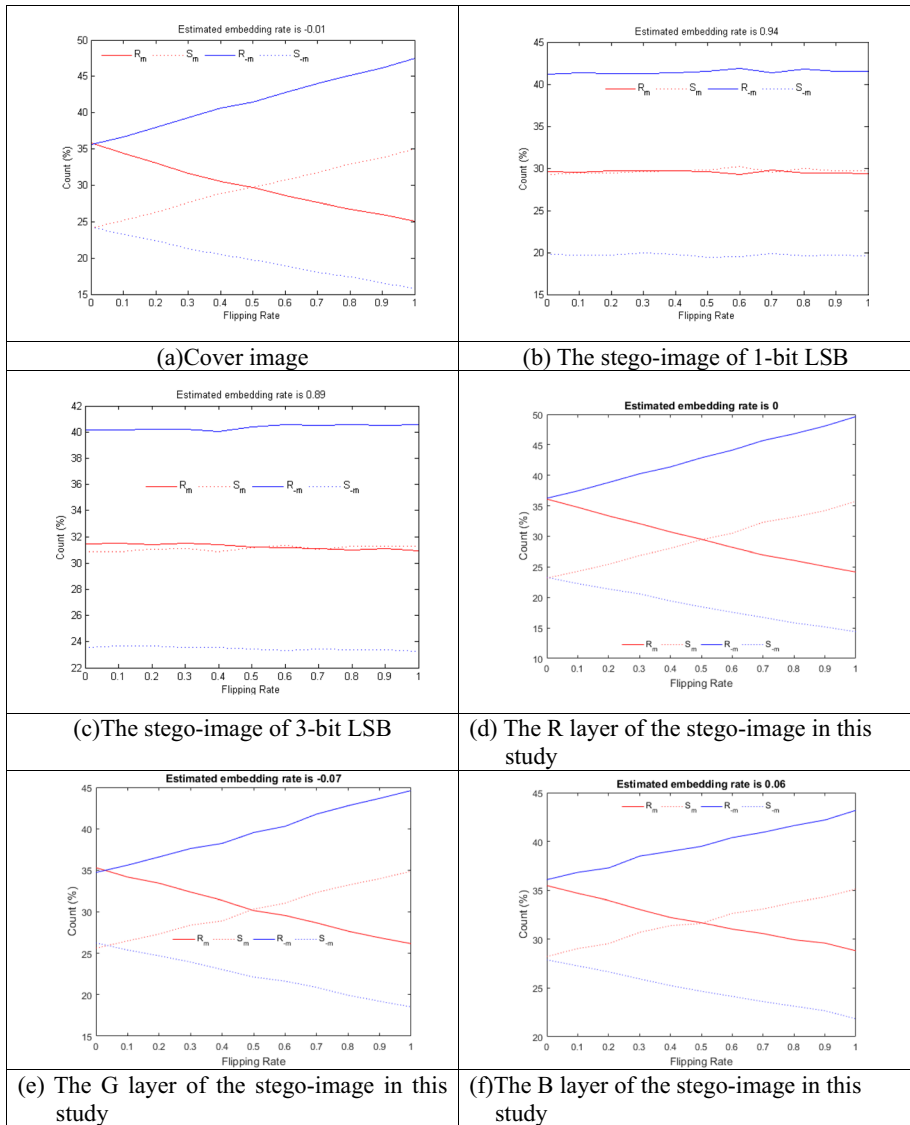


Fig. 9 Results of the RS Detection

a specific adjacent-pixel difference value appearing in an image is tallied [28]. Figure 10(a) displays the PDH of a regular image, with the horizontal and vertical axes representing the pixel difference values and their respective frequencies. A regular image typically exhibits a normal (Gaussian) distribution. However, a stego-image may present non-normal, stepped distributions, as depicted in Figs. 10(b), (c), and (d). These deviations from normal distribution serve as potential indicators for steganalysis, enabling the determination of whether an image contains concealed information.

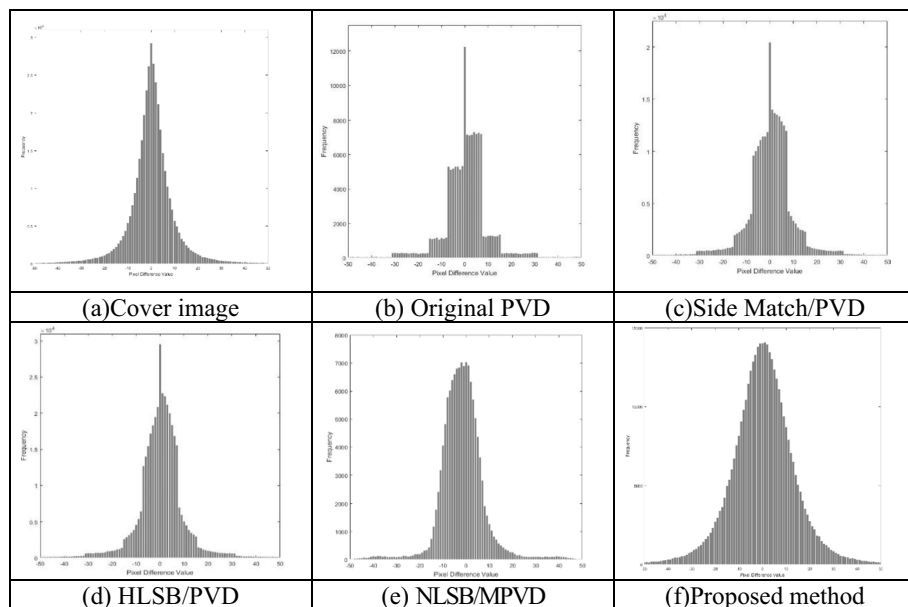


Fig. 10 PDH of pixel difference values with cover image and different steganographic methods

To address this, the present study conducts a security analysis on various techniques: the original PVD, the new Side Match/PVD, the mix PVD/LSB, the new LSB/PVD, and our own method. This analysis involves examining changes in the PDH features after embedding the cover image. Comparisons are made between the PDH of the cover image and that of the embedded image (refer to Fig. 10). The PDH of the cover image generally exhibits a distribution close to normal. However, as demonstrated in Figs. 10(b), (c), and (d), the height and width of the PDHs alter following embedding, resulting in non-normal distributions. Such modifications in PDH features following embedding offer opportunities for steganalysis.

In contrast, as shown in Fig. 10(f), the PDH generated by our method maintains a normal distribution, making it comparatively effective in evading steganalysis based on PDH features.

5 Conclusion

This study combined the characteristic of color space and edge detection techniques to propose a steganography technique with high concealing capacity while maintaining the quality of camouflaged images. During the experiment, the different sizes of blocks, 4×4 , 8×8 , 16×16 , and 32×32 were used to determine edge blocks. The amount of confidential information hidden and the quality of camouflage images of Lena, Baboon, and Airplane color carrier images were compared. The experimental results demonstrated that the data obtained by 32×32 blocks were the best. To verify the applicability of this method to various types of images, this study conducted extensive experiments with 450 full-color lossless high-definition images of 512×512 pixels from the color image database. The experimental results also demonstrated that the data obtained by 32×32 blocks were the best. This study compared 32×32 blocks experimental data to other scholars' results.

The results of this study were compared with the steganography techniques proposed by scholars such as Wu & Tsai, Swain, Prasad & Pal, and Hassaballah. The experimental results demonstrated that the amount of confidential information concealed by the proposed method was superior to the experimental data of the above-mentioned scholars, and the acceptable camouflage image quality can be maintained. During the experiment, RS detection and pixel difference histogram analysis were also used to analyze the security of the camouflage images generated by this method. The analysis results indicated that RS detection and pixel difference histogram analysis could not effectively distinguish the camouflage images generated by this method. Therefore, the color image steganography technique which combined the RGB model and edge detection proposed in this study can resist RS detection and pixel difference histogram analysis. How to apply the color image steganography technique proposed in this study to a broader range of areas is an important future development direction for this research.

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Availability of data and materials The cover images and stego images used in the experiment, the code of the proposed steganography and related experimental data in this study can be disclosed. Please contact author for data requests.

Declarations

Competing interests This study is for academic research and submission purposes only. The authors in this study declare that they have no competing interests.

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Authors and Affiliations

Hsing-Han Liu¹  · Yu-Ting Su¹

✉ Hsing-Han Liu
liu.hansh@gmail.com

¹ National Defense University, No. 70, Sec. 2, Zhongyang N. Rd., Beitou District, Taipei City 11258, Taiwan